LESSON 6.1

Division Expressions with Fractions and Whole Numbers

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can explain what dividing with a whole number and a unit fraction means — how many groups fit, or how much is in one group — find the quotient with a model, and check it by multiplying.

WHY THIS MATTERS Agent Reyes finds a 3-foot strip of tape at the crime scene. The lab needs the tape cut into 1/4-foot sections for analysis. How many sections can be made? The answer cracks the code to the locked evidence vault.

REMEMBER

Division Expressions with Fractions and Whole Numbers

Dividing Whole Numbers & Fractions Dividing by a fraction asks one question: how many of that small piece fit inside the whole amount? For 3 ÷ 1/4, we ask how many 1/4-foot pieces fit into 3 feet.

1. Name the question: 3 ÷ 1/4 asks how many 1/4s fit in 3, and 1/2 ÷ 4 asks how much is in each of 4 equal shares.
2. Use the model: each whole holds 4 fourths, so 3 wholes hold 3 × 4 = 12 fourths, and 3 ÷ 1/4 =
3. Check by multiplying: 12 × 1/4 =
4. Shortcut: dividing by 1/4 gives the same answer as multiplying by its reciprocal, 4

Words: **Reciprocal *(Recíproco)*** **Quotient *(Cociente)*** **Dividend *(Dividendo)*** **Divisor *(Divisor)*** **Unit fraction *(Fracción unitaria)***

PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each statement: does dividing by the fraction give a result GREATER than the dividend, or LESS than the dividend?

Groups: **Quotient GREATER Than Dividend** **Quotient LESS Than Dividend**

_____ 5 ÷ 1/3

_____ 8 ÷ 1/2

_____ 6 ÷ 1/4

_____ 4 ÷ 2

_____ 10 ÷ 5

_____ 12 ÷ 3

HINT Work out each quotient first, then compare it with the dividend — the number being divided.

PART 2 Practice the skill

show every step

2 **Circle the letter of the best answer. Show how you know.**

What is 6 ÷ 1/5?

(A) 1
(B) 5/6
(C) 6/5
(D) 30

HINT 6 ÷ 1/5 asks: how many 1/5-size pieces fit into 6 wholes? Predict first — more or fewer than 6?

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

What is 4.8 ÷ 2?

(A) 0.24
(B) 2.4
(C) 2.6
(D) 24

SHOW YOUR WORK

PART 3 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

Agent Reyes has used 6 feet of evidence tape, and that is 1/2 of a full roll. How many feet of tape are on a full roll? (6 ÷ 1/2 asks how much is in ONE whole roll.)

(A) 3 feet
(B) 12 feet
(C) 6 1/2 feet
(D) 1/12 foot

HINT The 6 feet is only part of the roll. Which part?

SHOW YOUR WORK

PART 4 Explain and correct

reasoning

5 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake

1. **Problem** How many 1/3-foot pieces fit into 4 feet of tape? Find 4 ÷ 1/3.
2. **Student writes** Dividing makes the answer smaller, so 4 ÷ 1/3 = 4 ÷ 3 ≈ 1.3 pieces.
3. **Student's answer** About 1.3 pieces

Which step has the mistake? Explain what went wrong, then show the correct work.

> **HINT** How many 1/3-foot pieces fit inside just ONE foot?

CORRECT WORK

Correct answer: _____

6 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake

1. **Problem** How many 1/5-cup scoops fill a 6-cup container? Find 6 ÷ 1/5.
2. **Student writes** I will flip the first number instead of the divisor: 6 ÷ 1/5 = 1/6 × 5 = 5/6 scoop.
3. **Student's answer** 5/6 of a scoop

Which step has the mistake? Explain what went wrong, then show the correct work.

> **HINT** Which number do you flip when you divide — the dividend (6) or the divisor (1/5)?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Why does dividing by a unit fraction always give a number larger than the dividend?

SENTENCE STARTER Dividing by 1/3 means finding how many ___ fit into ___, so the quotient is ___ than the dividend.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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WHY THIS MATTERS Agent Reyes finds a 3-foot strip of tape at the crime scene. The lab needs the tape cut into 1/4-foot sections for analysis. How many sections can be made? The answer cracks the code to the locked evidence vault.

PART 1 Practice the skill

show every step

1 **Circle the letter of the best answer. Show how you know.**

What is 8.4 ÷ 0.7?

(A) 0.12 (B) 1.2
(C) 12 (D) 120

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

What is 14.4 ÷ 1.2?

(A) 0.12 (B) 1.2
(C) 12 (D) 120

SHOW YOUR WORK

3 Circle the letter of the best answer. Show how you know.

What is 3.78 ÷ 0.6?

(A) 0.063
(B) 0.63
(C) 6.3
(D) 63

SHOW YOUR WORK

PART 2 Apply it

real-world problems

4 Circle the letter of the best answer. Show how you know.

A detective has 6 yards of crime scene tape. She needs to mark off sections that are each 1/3 of a yard. She wants to know how many sections she can mark. 6 ÷ 1/3 = ?

(A) 2
(B) 3
(C) 6
(D) 18

SHOW YOUR WORK

5 Circle the letter of the best answer. Show how you know.

A detective has 6 yards of crime scene tape. She needs to mark off sections that are each 1/3 of a yard. She wants to know how many sections she can mark. Why is the answer bigger than 6?

(A) Because the pieces are smaller than a whole yard, so many of them fit
(B) Because dividing always makes numbers bigger
(C) Because 1/3 is negative
(D) Because 6 × 3 is multiplication, not division

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A detective has 6 yards of crime scene tape. She needs to mark off sections that are each 1/3 of a yard. She wants to know how many sections she can mark. If each section were 1/2 yard instead, how many sections?

(A) 3
(B) 12
(C) 18
(D) 24

Rewrite each fraction _____

Work it out _____

Simplify _____

EXPLAIN YOUR THINKING

Why does dividing by a unit fraction always give a number larger than the dividend?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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